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## EDUCATION

**Dartmouth College**, Hanover, NH

June 2028

*Bachelor of Arts, Engineering Sciences & Bachelor of Engineering, Mechanical Engineering*

**Green City: Sustainable Engineering in Berlin Foreign Study Program**, Berlin, Germany

Spring 2026

*Sustainable and environmental engineering through applied coursework, field study across Germany*

**Manhattan Center for Science & Mathematics**, New York, NY

June 2024

*Graduated Top 3%, Principal's Honor Roll, Advanced Science Research Program Leader, Varsity Track & Field Captain*

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## PROFESSIONAL EXPERIENCE & PROJECTS

**FreeBee**, Hanover, NH

Fall 2025-Present

*Founder - Independent Product Development*

- Designed a handheld applicator that treats Varroa mites, the parasite behind the majority of U.S. honeybee colony loss
- Replaced synthetic miticides with organic treatment, protecting colonies, honey purity, and pollinators vital to ⅓ of U.S. food supply
- Cut treatment from 5 minutes to 11 seconds per hive (30× faster), validated on live colonies at New Hampshire farms and apiaries
- Reduced cost to \$10 per hive/year versus \$28 for industry alternatives, addressing a U.S. market of 3.8 million managed colonies

**Truss Bridge Design**, Hanover, NH

Summer 2026

*Structural Design and Testing*

- Designed a birch-plywood truss, researching structural forms and sketching concepts into SAP2000 models
- Iterated and developed the design in SolidWorks under finite element analysis (FEA), then laser-cut and assembled the final bridge
- Tested to failure: the 0.6 kg (1.3 lb) truss held 10.38 kN (2,334 lbs); achieving a structural efficiency ratio of 1800x

**Lozorno Water Systems**, Lozorno, Slovakia

Spring 2026

*Urban Infrastructure Consulting*

- Co-authored Sustainable Systems Integration Analysis, a live infrastructure study commissioned by the mayor of Lozorno, Slovakia
- Calculated the drinking water system's supply ceiling and 20-year demand using data from municipal, regional, and national records
- Modeled future usage rates, diagnosing that population growth will break current supply within the decade
- Designed and costed three supply solutions, matching each to EU and Slovak infrastructure grants

**Iarnród Éireann (Irish Rail)**, Cork, Ireland

Summer 2025

*Engineering Intern*

- Worked within the Track Delivery Unit that organizes maintenance, renewal, and construction across the southern rail network
- Maintained the live works schedule, updating track jobs, dates, locations, crews, materials, and machinery
- Attended on-site coordination meetings across the network, from Cork's train yard to a national session outside Dublin

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## LEADERSHIP EXPERIENCE

**Dartmouth Outing Club:** Co-led 2 expeditions of 12 across the U.S., managing budgets, permits, gear, food, travel, and safety

**Flora & Fauna Club:** Lead ecology-focused outings connecting students to local ecosystems and conservation efforts

**Dartmouth Orientation Team:** First-Year Trip Leader in student-run program, mentoring students as their first connection to campus

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## HONORS AND AWARDS

- Awarded Ethics Institute Sayles Research Grant to independently develop and field test an integrated, durable device for daily use
- Dartmouth Emerging Leader, Class of 2028: faculty-nominated among the top 5% of first-years for embodying the college's ideals
- Recognized as USATF All-American, USATF NY XC Champion, and USATF NY Athlete of the Year (17-18 Boys)
- Competed Internationally at Irish U20 Nationals (4th, 3000m Steeplechase), and a World Athletics Continental Tour Bronze meet

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## SKILLS & INTERESTS

**Design & Electronics:** SolidWorks, Fusion 360, Arduino, Raspberry Pi, ESP32, sensor integration

**Fabrication:** 3D printing & printer modification, machining, laser cutting, bike mechanics, prototyping, custom PC building

**Programming & Modeling:** C, Python, OpenRocket, MATLAB; data logging and modeling from coursework and projects

**Certifications:** Heartsaver First Aid, CPR & AED (AHA); Personal Track Safety (Iarnród Éireann)

**Volunteering:** Achilles International, God's Love We Deliver

**Interests:** poetry, lego, rocketry, Shakespeare, triathlon